

LEE HONG TENG

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EDUCATION

Singapore Management University

Bachelor of Science (Information Systems)

Aug 2021 – Jul 2025

- Major in Business Analytics
- Relevant Modules: Machine Learning & Applications, Data Mining & Business Analytics, Visual Analytics for Business Intelligence, Web Application Development, Enterprise Solution Development, Digital Business & Transformation

Zhejiang University (Semester Exchange)

School of Public Affairs

Feb 2024 – Jul 2024

- Courses taken: International Law and Relations, China Public Finance, Social Science and Cultural Hegemony, Consumer Neuroscience, Technology Entrepreneurship, Academic Writing in Psychology

Ngee Ann Polytechnic

Diploma in Business Information Technology

Jan 2016 – May 2019

- Awarded Tai Sin Prize for Most Outstanding Performance during '6-Month Internship' (GPA 3.59 / 4.0)

EXPERIENCE

Housing & Development Board (HDB)

Systems Analyst

Dec 2025 – Present

- **Enterprise Digital Transformation:** Led the AI & digital innovation pillar of the group's transformation strategy. Designed a three-phase operating model (Idea Capture; Assessment & Prioritisation; Development, Monitoring & Deployment) and built a productivity-impact model from scratch (four-factor savings formula with project tiering and compounded year-on-year savings)
- **Project BEACON:** Independently conceptualised and built an AI sense-making pipeline (POC → roadmap) to manage high volumes of resident cases. Features include: semantic case clustering via embeddings, AI-assisted escalation detecting sentiment/priority mismatches, and LLM-generated strategic analysis reports; ran stakeholder demos and initiated handover to the data science team for production
- **Staff SMS Audit Platform:** Evolved an AI-powered SMS compliance and quality-grading tool from a ~750-line Streamlit prototype into a Flask (JSON/SSE) + custom single-page platform with live streaming and org-level (Zone → Limb → Staff) analytics and PDF/Excel reporting
 - Rewrote the compliance rule engine and AI grading (each message scored 1–5 on Clarity, Tone & Grammar with sentiment and justification) across multiple LLM providers (GovTech gateway, OpenAI, Gemini)
 - Built emergent category discovery via a BERTopic pipeline (embeddings → UMAP → HDBSCAN → c-TF-IDF); engineered checkpointing to survive ~30s request limits and scaled clustering to ~250k-row uploads
- **Additional:** Framed use cases for internal assistant bots built on WOG tools and presented a Digital Transformation Study distilling a P.R.I.M.E. framework & Digital Maturity Assessment model, and an AI strategy briefing on UOB
- Technologies: Python, Flask, LLM APIs, RAG, Embeddings, BERTopic / UMAP / HDBSCAN, HTML/CSS/JS

Urban Redevelopment Authority

Systems Analyst (Contract)

Jul 2025 – Nov 2025

- Built an end-to-end Retrieval-Augmented Generation (RAG) pipeline powering a domain-specific AI chatbot over URA web and PDF content — automated scraping (Puppeteer, cron-based incremental/full refresh) and multi-modal PDF extraction (PyMuPDF)
- Processed content into embeddings stored in PostgreSQL + pgvector; implemented hybrid retrieval (vector similarity + BM25 keyword) with query expansion and cross-encoder reranking, plus a real-time monitoring dashboard
- Assisted in the development of an AI-powered Smart Compliance tool automating occupant identification and use-regulation checks with risk-based enforcement prioritisation (bulk CSV + single-entry), and a multi-step deep research / web-search agent (Exa Research API)

- Technologies used: Python, Next.js, TypeScript, PostgreSQL/pgvector, Puppeteer, Exa & SERP API

Urban Redevelopment Authority

Product Development Intern (AI/ML)

Jan 2025 – May 2025

- Developed the front-end of an in-house GPT chatbot powered by Gemini 2.5 Pro; ran comparative testing across Gemini 2.0/2.5 Flash, OpenAI o3/o3-mini, and GPT-4.1 with long-context and web-grounding evaluation
- Built a full-stack Notion-like email editor web app with GPT-4.1 AI features that drafts emails using context from the top related past emails and FAQs
- Technologies used: Prompt engineering, Next.js, HTML, TypeScript · [Internship Presentation](#)

Twimbit Pte Ltd

Research Intern

May 2023 – Jul 2023

- Researched and co-authored articles on ChatGPT in Financial Services, Buy Now Pay Later (BNPL), and Customer Experience in Singapore Banks ([published article](#))
- Supported conversion of slides & presentations into blog content; tested Twimbit's ChatGPT-powered website chatbot

PROJECTS

Final Year Project — Urban Forms Gen AI

Aug 2024 – Dec 2024

- R&D on GenAI solutions to generate building blocks for a given plot of land for URA; performed classification on building polygons and conducted UAT with end users

California Fires Dashboard

Jan 2025 – Apr 2025

- Built a web dashboard visualising California fire data — Tableau for initial exploration, front-end with D3.js & React, Node.js backend, containerised with Docker

Email Editor Web Application

Mar 2025 – May 2025

- Web app using Gemini Flash 2.0 to generate email drafts from received emails and embedded FAQ data (OpenAI text-embedding); built with Next.js & TypeScript, hosted on Vercel

Spreadsheets Analysis on Kangiiten

Jan 2023 – Apr 2023

- Data cleaning, exploration and visualisation on Shopify data; time-series analysis to forecast demand and optimise resource allocation

SKILLS

- Languages:** Python, Java, PHP, JavaScript/TypeScript (React, Vue, Next.js, D3), SQL, HTML, CSS
- AI / ML:** LLM Integration (GPT · Gemini · Claude), RAG Pipelines, Prompt Engineering, Embeddings & Vector Search, Topic Modelling (BERTopic · UMAP · HDBSCAN)
- Data & Infrastructure:** PostgreSQL/pgvector, Flask, Node.js, Docker, Git, MongoDB, Firebase, Vercel, Tableau, Power BI
- Tools:** Jira, Figma, Puppeteer, Exa API
- Certifications:** Java Certified Foundations Associate; AI100 Python Programming & Data Visualisation (Heicoders); AI200 Applied Machine Learning (Heicoders)

COMMUNITY SERVICE

Cahaya Community — SMU

Aug 2021 – Dec 2023

Mentor

- Mentored primary and secondary school students; assisted with homework and provided free tuition

SMU Rotaract Club — SMU

Aug 2021 – Jun 2022

Mentor

- Interacted with beneficiaries virtually and assisted in conducting meet-up sessions